LEVEL 19 · PART IX

Architecture

19

Goal: Design complete production systems and evaluate trade-offs like a senior engineer.

RECOMMENDED TIME	TOPICS
Ongoing	11

PRACTICE / CAPSTONE

Design (diagram + written ADRs + cost + DR plan) the reference architecture: CDN → Load Balancer → Kubernetes → API/Worker/Web → Redis → Message Queue → PostgreSQL/S3, with Prometheus, Grafana, Loki, OpenTelemetry and Sentry.

Topics

#	Topic	What you will learn
1	Scalability **p.3553**	scaling axes, stateless services, caching, queues, bottleneck analysis
2	Availability **p.3565**	SPOFs, multi-AZ/region, nines math, graceful degradation
3	Security Architecture **p.3577**	defense in depth, zero trust, threat modeling (STRIDE), blast radius
4	Performance **p.3589**	latency budgets, caching layers, profiling, load testing, tail latency
5	Cost **p.3601**	cost as a design constraint, unit economics, cost-aware design
6	Operational Complexity **p.3611**	boring technology, cognitive load, build vs buy, when not to use Kubernetes
7	Deployment Strategy **p.3623**	environments, release strategies, multi-region rollout, blast-radius control
8	Monitoring **p.3635**	designing observability in, SLO-driven alerting, telemetry pipeline cost
9	Disaster Recovery **p.3649**	choosing a DR tier per component, RPO/RTO driven design
10	Reference Production Architecture Walkthrough **p.3661**	CDN → LB → K8s → API/Worker/Web → Redis → queue → Postgres/S3 + observability stack

#	Topic	What you will learn
11	Trade-offs and Architecture Decision Records **p.3677**	writing ADRs, trade-off matrices, design reviews, estimating
★	Level capstone project **p.3689**	Design (diagram + written ADRs + cost + DR plan) the reference architecture: CDN → Load Balancer → Kubernetes → API/Worker/Web → Redis → Message Queue → PostgreSQL/S3, with Prometheus, Grafana, Loki, OpenTelemetry and Sentry.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

